User Testing Report:
Bridging Gaps: AI for Diagram Accessibility

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1 Goals for User Testing

The purpose of user testing in the Reading4All system is to evaluate the effectiveness, usability, and accessibility of the AI-generated alternative text for physics diagrams. Since the system is designed to support users with visual impairments, particularly those relying on screen readers, it is essential to ensure that the generated descriptions are clear, accurate, and meaningful in conveying the content of complex physics diagrams.

The primary goal of user testing is to assess whether users can successfully interact with the system and complete key tasks, such as uploading images, generating alt text, and reviewing or editing the output. This includes evaluating both functional performance and accessibility features.

In addition to the generated alt text, user testing also evaluates the user interface to ensure it is fully accessible. This includes verifying that the system is compatible with screen readers, supports effective keyboard navigation, and that all interactive elements are reachable and usable without a mouse. Furthermore, the interface is assessed against WCAG 2.1 Level AA guidelines to ensure it meets standard accessibility requirements.

Another key objective is to evaluate the quality of the generated alt text, specifically whether it provides sufficient detail, maintains clarity, and supports user understanding of physics concepts. This involves identifying issues such as unclear descriptions, missing information, or incorrect interpretations of diagrams.

Additionally, user testing aims to gather feedback on the overall user experience, including interface usability, workflow efficiency, and any barriers encountered during interaction. The insights gained from testing are used to identify areas for improvement and guide refinements to both the AI model and system design.

Overall, user testing ensures that the Reading4All system meets its goal of providing accessible, high-quality alternative text that enhances learning and accessibility for users engaging with physics diagrams.

2 List of User Tasks and Evaluation Rationale

User testing was conducted through two structured test scenarios: a functional workflow test that focused on screen reader compatibility and a keyboard navigation accessibility test. These tests were designed to evaluate the usability and accessibility of the `Reading4All` system, with a focus on ensuring that users with visual impairments can seamlessly navigate the interface and access all features without any barriers.

The tasks for both types of testing were outlined step by step in an Excel file so that testers had a structured procedure to follow and could easily record any comments/feedback for each step. For each test case, the tester documented the actual observation and had the option to include screenshots for any failed

test. This allowed us to collect data in a consistent manner and made it easier to analyze the results. Figure 1 shows an example of the Excel testing sheet used.

These test cases and their wording were developed and reviewed with guidance from our supervisor, who has extensive experience conducting user testing, particularly with visually impaired individuals. This helped us ensure that the instructions were clear, accessible and appropriate for one of our main target user groups. Additionally, the structure of the test was designed to balance data collection and ease of use, ensuring that testers can complete the tasks in a convenient and accessible manner.

Finally, users were asked to evaluate the quality of the generated alt text by answering two follow-up questions.

SCREEN READER TEST

Step	Test Case ID	Test Steps	Expected Results	Actual Observation	Status (PASS/FAIL)	Optional: Screenshots of Failed Steps
1	TC-SR01	Open web browser and navigate to Reading4All	Homepage loads successfully without errors		Pass	
2	TC-SR02	Start sign up process	User is redirected to Sign up		Pass	
4	TC-SR04	User retrieves code from email Inputs code into field Presses Sign Up	User is authenticated and registered		Pass	
5	TC-SR05	Start login process	User is redirected to Login page		Pass	
6	TC-SR06	Enter valid email and password	User is authenticated and redirected to Upload page		Pass	
7	TC-SR07	Upload valid image file (<2MB, supported format e.g., JPG/PNG)	Image upload is successful Image preview is displayed clearly "Generate Alt Text" button becomes visible and clickable		Pass	

Figure 1: Example of Excel testing sheet used during user testing.

2.1 Functional and Screen Reader Testing

Users were first asked to complete a full end-to-end interaction with the system while using their screen reader. This test aimed to evaluate both the functionality and compatibility with screen readers.

The tasks began with navigating today the `Reading4All` homepage and completing the sign-up process. This included entering a valid email, password and receiving a verification code to confirm their account was registered. Users then logged into the system with their credentials. Users were then asked to upload a valid image file and verify that the image preview was displayed correctly. They then began the alt text generation process and observed system feedback such as loading and generation button becoming disabled. After generation, users were instructed to review, edit and save the alt text. They also tested copying the generated text to the clipboard and pasting it into an external application such as `Word`. Next, users navigated to the history page to confirm that the previously generated entries and their edits were displayed correctly and in the expected order. They also tested copying alt text directly from this page. Finally, users logged out of the system and confirmed they were redirected to the homepage.

Ultimately, these testcases allowed us to determine if users could successfully complete all key tasks using as screen reader and to identify and barriers in navigation and system feedback.

2.2 Keyboard Navigation Testing

Users were also asked to evaluate the system using only keyboard navigation. Furthermore, users were asked to navigate to the homepage using the Tab key to ensure all interactive elements are reachable. They then completed the login process, uploaded an image, generated alt text, edited the text and copied it to their clipboard using only keyboard inputs. Users also verified that key interface elements such as buttons and input fields could be operated without a mouse. Finally, users tested the interface at 200% zoom to ensure the layout remained readable and did not break.

Table 1: Revision 0 Internal Evaluation Scores per Evaluator

Evaluator	**Sufficiency (1–3)**	**Length (1–3)**	**Accessibility (0–3)**	**Learning (0–3)**
Person 1	2	3	3	1
Person 2	2	3	3	2
Person 3	2	3	3	1
Person 4	2	2	3	1
Person 5	2	2	3	1
Mean	**2.0**	**2.6**	**3.0**	**1.2**
Acceptable	≥ 3	≥ 3	≥ 2	≥ 2

2.3 Alternative Text Evaluation

To evaluate the quality of the generated alternative text, users were asked two questions after completing their testing. These questions were designed to assess the usefulness and quality of the generated descriptions. The first question, `"Does the generated alt text convey sufficient information to achieve the stated objective?"` aims to determine whether the description purposed a sufficient amount of details for the user to understand the purpose and meaning of the image. The second question, `"Is the alt text detailed enough for understanding the image?"` helped us assess whether the generated text goes beyond a general summary and focuses on the technical details required when learning physics.

These questions were chosen in order to gather feedback from users about on any missing aspects from the generated text and to identify any key issues.

3 Internal Development Testing

Prior to external user testing, the development team — Nawaal Fatima, Dhruv Sardana, Fiza Sehar, Moly Mikhail, and Casey Francine Bulaclac — conducted informal internal evaluations of the generated alternative text across two revision cycles. The diagrams used for internal testing were the same open-source images drawn from Grade 12 physics textbooks that were later used in external user testing.

Each team member independently reviewed the generated alt text and rated it across four metrics: Sufficiency (1–3), Length (1–3), Accessibility (0–3), and Learning (0–3), where Sufficiency measures whether the description conveys enough information, Length assesses whether the output is an appropriate level of detail, Accessibility evaluates how well the text supports users relying on screen readers, and Learning captures whether the description supports understanding of physics concepts.

3.1 Revision 0 Results

Table 1 presents the internal evaluation scores collected from five evaluators for the Revision 0 model outputs.

Revision 0 results showed that while Accessibility consistently met the acceptable threshold across all evaluators (mean: 3.0), both Sufficiency (mean: 2.0) and Learning (mean: 1.2) fell below their respective thresholds of ≥ 3 and ≥ 2. Length (mean: 2.6) narrowly missed its threshold. These results indicated that while the system produced accessible output, the generated descriptions lacked the technical depth and completeness needed to support physics learning, motivating further refinement of the model and prompting strategy.

3.2 Revision 1 Results

Table 2 presents the internal evaluation scores for the Revision 1 model outputs, collected using the same rubric and evaluator pool as Revision 0.

Revision 1 showed measurable improvement across all metrics. Length reached its acceptable threshold (mean: 3.0), and both Sufficiency (mean: 2.4) and Learning (mean: 1.8) improved notably from Revision 0, though neither fully met their thresholds. Accessibility remained consistently strong (mean: 3.0). These

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Person 1	2	3	3	2
Person 2	3	3	3	2
Person 3	2	3	3	2
Person 4	2	3	3	1
Person 5	3	3	3	2
Mean	**2.4**	**3.0**	**3.0**	**1.8**
Acceptable	≥ 3	≥ 3	≥ 2	≥ 2

results suggest that the prompt refinements made between revisions had a positive effect, particularly in producing more complete and appropriately detailed descriptions. However, the Learning metric indicates that further work is needed to ensure the generated text adequately supports physics comprehension.

4 Results Summary

4.1 Positive Findings

The screen reader testing showed that users were able to navigate through the full workflow without errors. All key pages loaded correctly, and users could successfully sign up, log in, upload images, generate alternative text, edit results, and copy text to the clipboard. The system also allowed users to access previously generated alt text through the history page. These results indicate that the application supports end-to-end task completion using assistive technologies.

Keyboard navigation testing further confirmed strong accessibility support. All interactive elements were reachable using the Tab key, and buttons could be activated using Enter or Space. Users were able to upload images, generate alt text, edit content, copy results, and navigate between pages without using a mouse. The interface remained usable and readable at 200% zoom, demonstrating responsiveness and compatibility with accessibility guidelines. Overall, the keyboard accessibility of the system was effective and consistent.

4.2 Issues Identified

Despite the positive results, several usability issues were observed. During login and alt text generation, there were occasional delays where actions required multiple attempts. This behavior was likely due to backend services waking from inactivity, but it impacted perceived responsiveness.

Some accessibility feedback indicated uncertainty regarding the visibility of focus indicators when navigating using the keyboard. Although navigation was possible, clearer visual focus indicators would improve usability and confidence for keyboard users.

The generated alternative text quality also revealed areas for improvement. In some cases, the descriptions included incorrect or missing information, lacked clarity, or did not fully describe spatial relationships between elements. Certain outputs were fragmented and not written in complete sentences, making them harder to understand. Additionally, important diagram components such as sections, symbols, or spectrum bars were sometimes omitted.

At higher zoom levels (400%), layout overlap issues were observed, particularly with header content. This indicates that while the interface performs well at 200% zoom, further adjustments are needed to fully support extreme zoom accessibility requirements.

4.3 Summary

In summary, the system performed well in terms of accessibility and task completion using both screen readers and keyboard navigation. However, improvements are needed in system responsiveness, focus indicator visibility, alt text accuracy and completeness, and layout behavior at higher zoom levels.

5 Analysis and Insights

The results from user testing indicate that users were generally able to complete the primary workflow successfully, including uploading images, generating alternative text, and navigating through the system. This suggests that the core functionality of the Reading4All system is effective and that the interface supports task completion. The positive results related to screen reader compatibility and keyboard navigation demonstrate that the system meets its accessibility goals. Users were able to interact with most interface components without requiring a mouse, indicating good compliance with accessibility guidelines.

However, the results also highlighted several usability issues. Some users experienced difficulty identifying certain buttons and interactive elements, which caused hesitation during navigation. This suggests that visual cues and button styling should be improved to make actions more intuitive. Additionally, feedback from alt text evaluation showed that while descriptions were generally helpful, some outputs lacked sufficient technical detail for complex physics diagrams. This indicates that further improvements to the model or prompting strategy may be required. Overall, the testing results show that the system is functional and accessible, but refinements to interface clarity and alt text detail will improve usability and better support user needs.